

INDIAN STATISTICAL INSTITUTE, KOLKATA
FINAL EXAMINATION: FIRST SEMESTER 2024 -'25
M. MATH I

Subject: Linear Algebra Date: November 25, 2024
Duration: 4 hrs Maximum score: 70

Attempt all the problems. Justify every step in order to get full credit of your answers. All arguments should be clearly mentioned on the answerscript. Points will be deducted for missing or incomplete arguments.

In this question paper, assume all vector spaces have finite dimension and characteristic of any field is NOT 2.

- (1) Let $T \in L(V, V)$ be simple (that is, the only T -invariant subspaces of V are the two trivial ones).

(a) If $S \in L(V, V)$ such that $ST = TS$, then show that S is either zero or an isomorphism.

Hints: Establish injectivity of S .

(b) Show that for all $v_1, v_2 \in V \setminus \{0\}$, there exists unique $S \in L(V, V)$ such that $sv_1 = v_2$ and $ST = TS$.

Hints: Work with polynomials in T .

(c) Show that $\dim \{S \in L(V, V) : ST = TS\} = \dim(V)$. [12 marks]

- (2) (a) Let X be an invertible complex $n \times n$ -matrix such that X^5 is diagonalizable. Show that X is diagonalizable. Will the conclusion continue to hold without the assumption of invertibility of X ? Justify.

Hints: Start working with the Jordan form and minimal polynomial of X .

(b) Let A, B be invertible complex $n \times n$ -matrices such that $BA^2B = A^3$ and B has n -many distinct eigen values. Show that A is diagonalizable.

Hints: First show that A^5 and B commute with each other. [12 marks]

- (3) Let V be a finite dimensional vector space over $\mathbb{F}$, $T \in L(V, V)$ and $p(x) \in \mathbb{F}[x]$ be monic irreducible such that V has a cyclic vector for T and $\chi_T(x) = [p(x)]^4$. Find a T -invariant subspace W of V such that

(a) $T|_W$ is simple,

(b) there does not exist any T -invariant subspace of V which is complementary to W .

Justify.

Hints for part (b): Suppose the statement is false. Consider the minimal polynomial and arrive at a contradiction. [11 marks]

- (4) Let V be a finite dimensional complex inner product space and $T \in L(V, V)$.

(a) If T is normal, then show that dimensions of the subspaces $\text{Range}(T)$ and $\text{Range}(T^*)$ are identical.

(b) Show that T is normal if and only if there exist unitary $U \in L(V, V)$ such that $T^* = UT$. [12 marks]

(5) Let V, W be a finite dimensional complex inner product space.

(a) If $T \in L(V, W)$, then show that $\ker(T) = \ker(T^*T) = \text{Range}(T^*)^\perp$.

(b) Show that $\left(\text{span} \bigcup_{j=1}^k S_j\right)^\perp = \bigcap_{j=1}^k S_j^\perp$ for all $k \in \mathbb{N}$ and $S_1, \dots, S_k \subset V$.

(c) If $T_1, \dots, T_k \in L(V, W)$ are positive, then show that $\sum_{j=1}^k T_j$ is also positive and

$$\text{span} \bigcup_{m=1}^k \text{Range}(T_m) = \text{Range}\left(\sum_{j=1}^k T_j\right). \quad [11 \text{ marks}]$$

(6) (a) Let A, B be two idempotents. Show that $A + B$ is idempotent if and only if $AB = 0 = BA$.

(b) Let P, Q be two orthogonal projections such that $\text{Range}(P) \subset \text{Range}(Q)$. Show that $Q - P$ is an orthogonal projection.

(c) Let $P_1, \dots, P_k$ be orthogonal projections such that $\sum_{j=1}^k P_j$ is also orthogonal projection. Show that $\text{Range}(P_j) \perp \text{Range}(P_m)$ for all $j \neq m$.

Hints: Mathematical induction on k and use (5)(a), (5)(c) and (6)(b).

[12 marks]

(7) (a) Let $N_1, N_2 \in L(V, V)$ be nilpotent such that $N_1 N_2 = N_2 N_1$. Show that every element of $\{f(N_1, N_2) : f \in \mathbb{F}[x, y] \text{ such that } f(0, 0) = 0\}$ is nilpotent.

(b) If $T \in L(V, V)$ such that m_T is irreducible, then how are $\deg(m_T)$ and $\dim(V)$ related? Justify.

(c) Give an example of a square complex matrix which is not normal and diagonalizable. Justify.

Hints: Work with a basis which is not orthonormal. Turn it into an eigen basis.

[11 marks]

$$\begin{aligned} (Q-P)^2 &= Q^2 - PQ - QP + P^2 = Q - PQ - QP + P \\ (Q-P)^* &= Q^* - P^* = Q - P \quad Q(v) = P(Q(v)) \\ (A+B)(A-B) &= A^2 - B^2 + AB - BA \\ P(Qv) &= Qv = A^2 - B^2 + AB - BA \end{aligned}$$